

LESSON 3-7 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Convert Between Systems

Builds on Lesson 3-7 Session 1 • 6.RP.A.3d

TODAY'S LESSON

You will change a 66 lb dog to kilograms before you find the dose.

YOU NEED TOConvert between systems, like $20 \text{ kg} \approx 44 \text{ lb}$. **You will use it in Practice Form A, problems 2 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** the fact, like $1 \text{ kg} \approx 2.2 \text{ lb}$.
- 2** New unit smaller? **Multiply**. New unit larger? Divide.
- 3 Write** the answer with the new unit.

WATCH ME*the steps in action***A suitcase is 20 kg. About how many pounds is that?****Step 1** $1 \text{ kg} \approx 2.2 \text{ lb}$.**Step 2** A pound is smaller, so multiply: $20 \times 2.2 = 44$.**Step 3** $20 \text{ kg} \approx 44 \text{ lb}$.**Answer about 44 lb****TRY IT WITH HELP***fill in each step*

- 1.**
- About how many kilograms is 22 lb?

Step 1 Fact: $1 \text{ kg} \approx$ _____ lb.**Step 2** A kilogram is larger, so divide: $22 \div 2.2 =$ _____.**Step 3** $22 \text{ lb} \approx$ _____ kg.

- 2.**
- About how many kilometers is 5 miles?

Step 1 Fact: $1 \text{ mi} \approx$ _____ km.**Step 2** A kilometer is smaller, so multiply: $5 \times 1.6 =$ _____.**Step 3** $5 \text{ mi} \approx$ _____ km.**YOUR TURN***use the steps on your own*

- 3.**
- Convert. Use
- $1 \text{ kg} \approx 2.2 \text{ lb}$
- and
- $1 \text{ mi} \approx 1.6 \text{ km}$
- .

a. $5 \text{ kg} \approx$ _____ lb**b.** $30 \text{ kg} \approx$ _____ lb**c.** $55 \text{ lb} \approx$ _____ kg**d.** $6.6 \text{ lb} \approx$ _____ kg**e.** $10 \text{ mi} \approx$ _____ km**f.** $48 \text{ km} \approx$ _____ mi

Use a Rate per Unit

Builds on Lesson 3-2 Session 1 • 6.RP.A.2

TODAY'S LESSON

You will find a dose of 4 mg per kg for a dog that weighs 30 kg.

YOU NEED TOUse a rate per unit, like $5 \text{ mg per kg} \times 12 \text{ kg} = 60 \text{ mg}$. **You will use it in Practice Form A, problems 1 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Find the **rate**: the amount for 1 unit.
- 2** Find the **number** of units.
- 3** **Multiply**: number $\times$ rate.

WATCH ME*the steps in action***Plant food uses 3 grams per liter. How much goes in 12 liters?****Step 1** Rate: 3 g per liter.**Step 2** Number of liters: 12.**Step 3** $12 \times 3 = 36 \text{ g}$.**Answer** **36 grams****TRY IT WITH HELP***fill in each step***4.** A dose is 5 mg per kg. How much does a 16 kg dog need?**Step 1** Rate: _____ mg per kg.**Step 2** Number of kg: _____.**Step 3** $16 \times 5 =$ _____ mg.**5.** Take 2 tablets per day. How many tablets in 14 days?**Step 1** Rate: _____ tablets per day.**Step 2** Number of days: _____.**Step 3** $14 \times 2 =$ _____ tablets.**YOUR TURN***use the steps on your own***6.** Find each total.**a.** 3 cups per batch, 7 batches: _____**b.** \$4 per pound, 9 pounds: _____**c.** 8 g per liter, 2.5 liters: _____**d.** 6 mg per kg, 20 kg: _____**READY CHECK***a problem like the one in today's lesson***7.** At 5 mg per kg, how much medicine does a 9 kg cat need?**A** 14 mg**B** 1.8 mg**C** 4 mg**D** 45 mg**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #_____